Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

2 hours · 100 marks · 10–11 questions · ≈ 1.2 min per mark · calculator allowed, but "without sufficient working, correct answers may be awarded no marks".

ON THE FORMULA SHEET

sphere and cone mensuration · AP/GP sums, S_∞ · binomial series · quotient rule · cosine rule · sin, cos, tan (A ± B) · change of base

Not given: product and chain rules, standard derivatives/integrals, sector formulas, double angles, volume of revolution.

TOP 12 MARK-LOSERS

- "Use algebra / calculus / show working": answer alone = **0**.
- Wrong quadratic + calculator roots, no method = method mark lost.
- Integration step not shown = M0 M0 A0.
- Limits not substituted visibly when the answer is wrong.
- Rounded values in an **exact** answer.
- Degrees in calculus or sectors (radians only).
- Extra solutions in range; missing ones.
- Asymptote written as "−5" instead of **x = −5**.
- δx = 0.2 instead of 0.2% of x.
- Least n: inequality not reversed after ÷ ln(r < 1).
- Missing brackets that are never recovered.
- Line not in the form asked (integers, "= 0").

QUADRATICS AND POLYNOMIALS

MUST KNOW

$ax^2 + bx + c = a(x + b/2a)^2 + c - b^2/4a$; turning point $(-b/2a, c - b^2/4a)$

Discriminant: > 0 two roots · = 0 equal · < 0 none

$\alpha + \beta = -b/a$, $\alpha\beta = c/a$ · $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$ · $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

New equation: x^2 − (sum)x + product = 0 → integer coefficients

- $-3x^2 + 12x - 2 = -3(x - 2)^2 + 10$ (max 10, x = 2).
- $4x^2 - 3x - 8 = 0$, roots 1/α, 1/β → $8x^2 + 3x - 4 = 0$.
- Factor theorem: f(a) = 0. Remainder on ÷(x − a) = f(a).
- $6x^3 - 31x^2 + 25x + 12 = (x - 4)(3x + 1)(2x - 3)$.
- Line touches curve once: substitute, discriminant = 0.

LOGARITHMS

LAWS

$\log xy = \log x + \log y$ · $\log(x/y) = \log x - \log y$ · $k \log x = \log x^k$

$\log_a x = \log_b x / \log_b a$ · $\log_{a^2} x = \log_a x$

- One base → one log each side → remove logs → solve → **check arguments > 0**.
- $\log_7(8x^2 - 6x + 3) - \log_{49} x^2 = 3 \log_7 2 \Rightarrow 8x^2 - 14x + 3 = 0 \Rightarrow x = ¼, 3/2$.

SERIES

FORMULAE

AP: $u_n = a + (n - 1)d$ · $S_n = (n/2)[2a + (n - 1)d]$

GP: $u_n = ar^{n-1}$ · $S_n = a(1 - r^n)/(1 - r)$ · $S_\infty = a/(1 - r)$, $|r| < 1$

$\Sigma_{r=10}^{n} = \Sigma_{r=1}^{n} - \Sigma_{r=1}^{9}$

- Least n: $a + ar = 400$, $ar + ar^2 = 100 \Rightarrow r = ¼$, $a = 320$; $S_n > 426.6 \Rightarrow 0.25^n < 1.5625 \times 10^{-4} \Rightarrow n > 6.32 \Rightarrow$ **n = 7** (sign flips: ln 0.25 < 0).
- Two conditions → simultaneous equations in a, d (or a, r).
- Reject non-integer or negative n.

BINOMIAL

METHOD

$(a + bx)^n = a^n(1 + bx/a)^n$, valid $|x| < |a/b|$. Bracket bx/a fully.

$(1 - 4x)^{½} = 1 - 2x - 2x^2 - 4x^3$; x = 0.06 ⇒ $\sqrt{0.76} \approx 0.871936 \Rightarrow \sqrt{19} = 5\sqrt{0.76} \approx 4.360$.

- Coefficients as exact fractions in lowest terms.
- Approximation: x inside validity; show the substitution.
- Integrate the series term by term to estimate an area.

COORDINATE GEOMETRY

TOOLKIT

m = Δy/Δx · perpendicular $m_1 m_2 = -1$ · distance $\sqrt{(\Delta x^2 + \Delta y^2)}$

Ratio m : n on AB: A + (m/(m + n))(B − A)

Area: shoelace $½|\Sigma(x_i y_{i+1} - x_{i+1} y_i)|$ or split into triangles

- Point from distance + gradient: two equations → quadratic → use the given sign condition.
- Not a perpendicular bisector: the foot ≠ midpoint.

GRAPHS

- **Region R:** draw lines across the grid; shade or label as asked; test P = ax + by at each vertex.
- **Solve by graph:** rearrange so one side = plotted curve; draw the other side as a line; read x to 1 d.p.; leave the line visible.
- **y = (ax + b)/(cx + d):** asymptotes x = −d/c, y = a/c (as equations); intercepts; both branches; $dy/dx = (ad - bc)/(cx + d)^2$.
- No intersection with a line: discriminant < 0.

TRIGONOMETRY AND 3D

FORMULAE

$s = r\theta$ · sector $\frac{1}{2}r^2\theta$ · segment $\frac{1}{2}r^2(\theta - \sin\theta)$ (radians)

$a/\sin A = b/\sin B$ · $a^2 = b^2 + c^2 - 2bc\cos A$ · area $\frac{1}{2}ab\sin C$

$\sin 2A = 2\sin A\cos A$ · $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$ · $\tan 2A = 2\tan A/(1 - \tan^2 A)$

- Sector area 16.8, perimeter 16.4 ⇒ $r^2 - 8.2r + 16.8 = 0$ ⇒ r = 4, θ = 2.1.
- Obtuse angle: 180° − calculator value.
- Line–plane angle: line vs its projection. Plane–plane angle: lines ⊥ to the common edge.
- No solutions: sin θ = 5/3 or −3/2, outside [−1, 1] (say why).
- Widen the range for 2x − 75; all solutions, none extra.

VECTORS

- Everything in terms of a and b.
- Intersection: OX = λOB and OX = OM + μMN; equate coefficients.
- Collinear: scalar multiple + common point. Unit vector v/|v|.
- Area ratio: same height → ratio of bases.

DIFFERENTIATION

y	y′	y	y′
x^n	nx^{n-1}	e^{kx}	ke^{kx}
sin kx	k cos kx	ln x	1/x
cos kx	−k sin kx	uv	u′v + uv′
f(g(x))	f′(g)g′	u/v	$(u'v - uv')/v^2$

- Normal: $y = (x^3 - 2x)e^{1-x}$ at (1, −1): y′ = 2 ⇒ x + 2y + 1 = 0.
- Max/min: y′ = 0, justify with y″ (> 0 min, < 0 max).
- Optimise: constraint → one variable → differentiate → 0 → justify → substitute. $S = 2\pi r^2 + 180\pi/r$: $r^3 = 45$, r = 3.56, S = 238.
- Rates: dS/dt = (dS/dr)(dr/dt). Sphere: dV/dt = 3, r = 10 ⇒ dS/dt = 0.6.
- Small change: $\delta y \approx y'\delta x$; 2% ⇒ δx = 0.02x.

KINEMATICS

$a = dv/dt$ · $s = \int v\,dt + c$ · rest: v = 0 · distance: split where v = 0

$v = 3t^2 - 9t + 7$: disc. < 0 ⇒ never at rest; distance 0 → 5 = 47.5 m.

$v = 2(t - 3)(t - 5)$: $MN = |\int_3^5 v\,dt| = 8/3$ m.

INTEGRATION

RESULTS

$\int x^n = x^{n+1}/(n + 1)$ · $\int e^{kx} = e^{kx}/k$ · $\int \sin kx = -\cos kx/k$ · $\int \cos kx = \sin kx/k$

Area $\int y\,dx$ (split below axis) · between: ∫(upper − lower)

Volume: $\pi\int y^2\,dx$ (x-axis) · $\pi\int x^2\,dy$ (y-axis)

- $(x - 3)(x - 2)(x + 3)$, region 0 → 3: $\int_0^2 - \int_2^3 = 211/12$.
- $y = \sqrt{(4x - 8)}$: $\pi\int_2^b (4x - 8)\,dx = 2\pi(b - 2)^2 = 50\pi$ ⇒ b = 7.
- Limits from intersections, shown algebraically.
- Add/subtract a cone ($\frac{1}{3}\pi r^2 h$) for line parts.

FINAL 60 SECONDS

- Working shown? Exact or 3 s.f.? Degrees or radians?
- Every root checked; every extra rejected with a reason.
- Integrals: expression + both limits visible.